



F1-18-MKII Energy Losses in Pipes

The unit consists of a vertical test pipe on the side of the equipment which can be fed directly from the hydraulics bench supply or, alternatively, from the integral constant head tank above.

For large pressure differentials a digital handheld manometer is used.

PRODUCT CODE: F1-18-MKII

Categories: Educational, F1 Series Fluid Mechanics, F1-18 Energy Losses In Pipes

Tags: Armfield , Energy Losses In Pipes , F1-18 , Flow Rate , Fluid Mechanics , Hydraulically Smooth Circular Pipe , Hydraulics Bench , Laminar Flow , Large Pressure Differentials , Mercury Manometer , Pressure Differentials , Turbulent Flow Regimes , Water Manometer

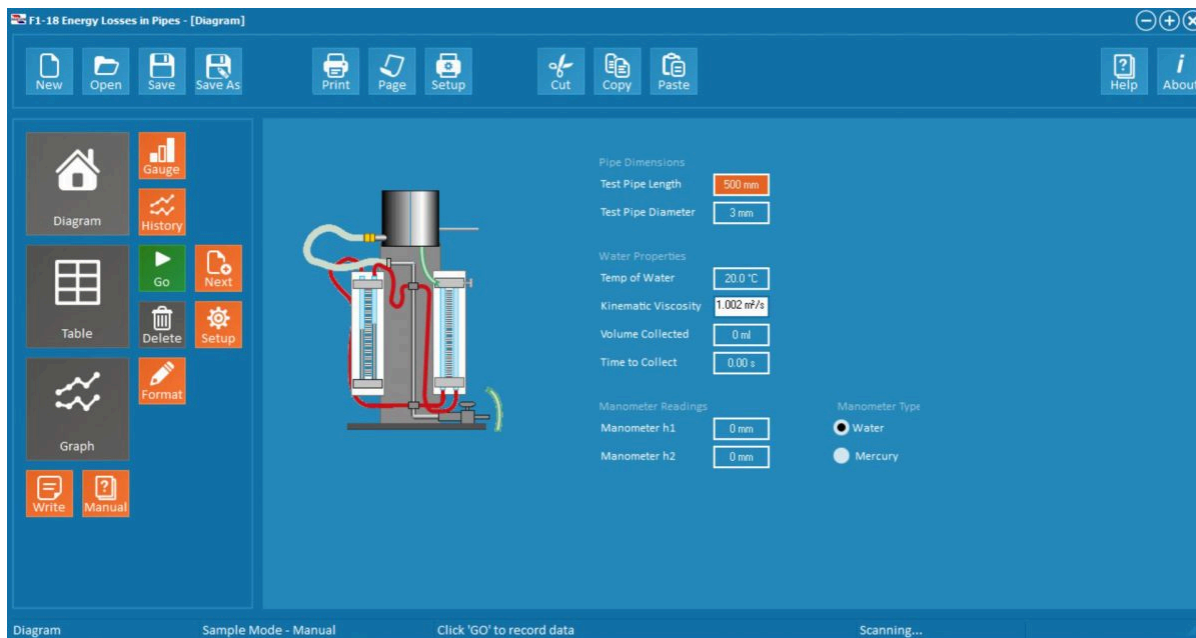
DESCRIPTION

The unit consists of a vertical test pipe on the front of the equipment which can be fed directly from the hydraulics bench supply or, alternatively, from the integral constant head tank above.

These in turn provide high or low flow rates which may be controlled by a valve at the discharge end of the test pipe. Manometers are used to measure the head loss.

For large pressure differentials a digital handheld manometer is used. In addition, a pressurised water manometer for small pressure differentials is also fitted to the unit.

Excess water discharging from the constant head tank is returned to the sump tank of the hydraulics bench.



Armfield F1-18-MKII Energy Losses in Pipes Software

TECHNICAL SPECIFICATIONS

Diameter of test pipe: 3.0mm

Length of test pipe: 674mm

Distance between pressure tapping points: 500mm

Range of digital manometer: 103 kPa

Range of water manometer: 500mm

Measuring cylinder capacity: 1000mm

Measuring jugs provided: 100ml / 250ml / 500ml

Requires Hydraulics Bench Service unit F1-10/F1-10-2

FEATURES & BENEFITS

Experimental Content

- To investigate the head loss due to friction in the flow of water through a pipe and to determine the associated friction factor over a range of flow rates in laminar flow
- To investigate the head loss due to friction in the flow of water through a pipe and to determine the associated friction factor over a range of flow rates in turbulent flow
- Determining the critical Reynolds number

Description



The Energy Losses in Pipes accessory consists of a test pipe, orientated vertically on the front of the equipment, which may be fed directly from the hydraulics bench supply or, alternatively, from the integral constant head tank.

These sources provide high or low flow rates which can be controlled by a valve at the discharge end of the test pipe. Head loss between two tapping points in the test pipe is measured using two manometers, digital hand-held manometer for large pressure differentials and a pressurised water manometer for small pressure differentials.

Excess water discharging from the constant head tank is returned to the sump tank of the hydraulics bench.

F1-18-MKII Videos (English and Español)

Requirements

F1-10 Hydraulics Bench



Dimensions

Length: 0.34m

Width: 0.48m

Height: 1.04m

Order Codes

F1-18-MKII

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